

4. Passing Primitive

Define a primitive function to let the vehicle **PASS** within selected **boundaries**.

- **Final Conditions:**

$a_f = 0 \rightarrow$ final acceleration equal to zero;

$v_f \in [v_{min}, v_{max}] \rightarrow$ final velocity between a min and a max value;

$s_f = \bar{s}_f \rightarrow$ final position equal to the position of the traffic light;

$t_f \in [T_{min}, T_{max}] \rightarrow$ final time between a min and a max value;

- **Initial Conditions:**

$v_0 = \bar{v}_0 \rightarrow$ initial velocity given by the sensor;

$a_0 = \bar{a}_0 \rightarrow$ initial acceleration given by the sensor;

Find the optimal **time** and **velocity** solution:

the total cost is derived w.r.t. v_f and imposed equal to zero $\rightarrow$ new function to define the optimal **velocity** needed to reach a given final position in a given time $\rightarrow$ new function to define the **time** to reach a final position and a final velocity.



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